Year 9, chapter 4.2: Sound

What is sound?

_____

_____

What is a sound wave?

_____

_____

Describe:

a) Compressions:
   _____

b) Rarefactions:
   _____

Explain why sound does not travel through outer space. (outer space is a vacuum = containing no particles).

_____

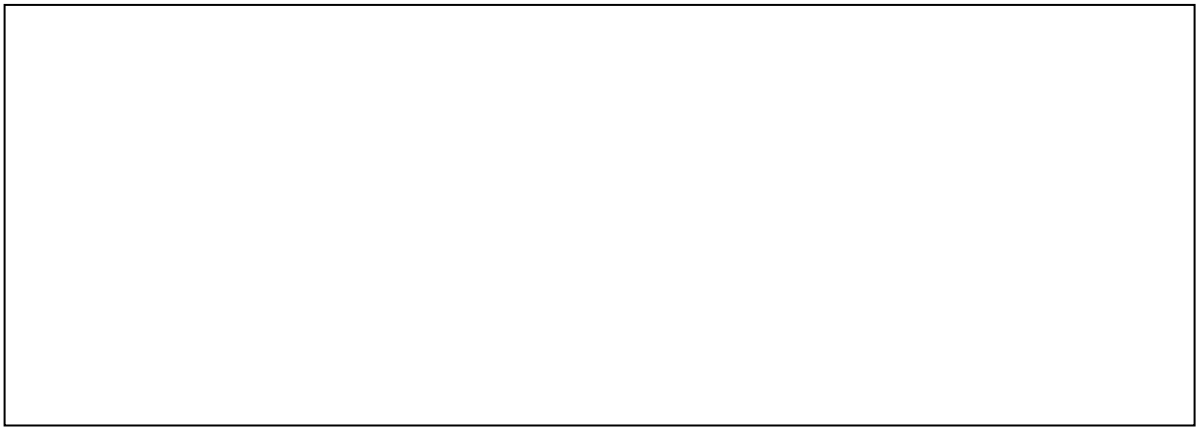
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Draw a particle diagram showing areas of compression and rarefaction of a sound wave produced by

a speaker. Label all parts.



What are the two types of waves?

1 -

_____

_____

(Diagram of wave type)

2-

_____

_____

(Diagram of wave type)

State the speed (in metres per second, m/s) that sound travels through:

Air (at $0^{o}C$): _____

Water: _____

Wood: _____

Steel: _____

Glass: _____

Explain why sound waves travel at different speeds though different media (substances)

_____

_____

_____

Sometimes when you shout or speak, especially in large open spaces, you can hear an echo of the sound produced. What is an echo and how is this produced?

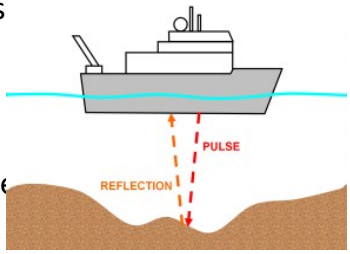
Sonar (sound, navigation and ranging technique) calculations

Sonar travels at about 1500m/s in sea water. Sonar measures the time it takes for sound to return to the ship. This data may be used to calculate how deep the water is, or how deep a school of fish are:

To calculate depth:

1. Have the time it takes for the sonar signal to return. This gives the time it takes for the signal to reach the bottom or the fish.
2. Multiply this by the speed of sound in sea water (x 1500 m/s

Depth (m) = speed (m/s) x



Calculate the following depths assuming speed of sound in sea wate
m/s.

It took 0.4s for a sound pulse from a ship to return. What is the depth of the

water?

Calculate the depth of the fish, given that the sound took 1 second to return.

Think about it!

When there is a storm, why do you always see the lightning before you hear the ‘boom’ of the thunder?

_____

_____

Fighter jets can create sonic booms that are capable of smashing windows! Briefly describe how this can happen.

_____

_____

_____

_____

Why are ‘sound rooms’ where bands often practise covered with bumpy, soft cushioned material?

_____

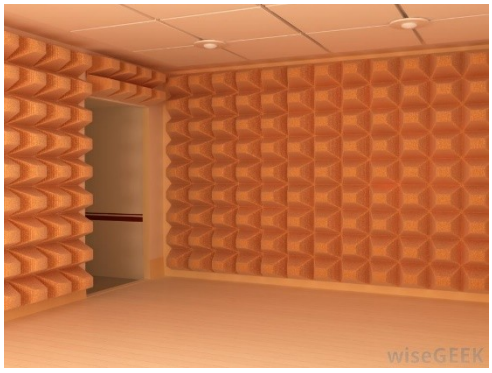
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Sound proof room material

Frequency and Pitch

What does a cathode ray oscilloscope (CRO) measure?

_____

_____

What is frequency?

_____

_____

What is frequency measured in?

_____

_____

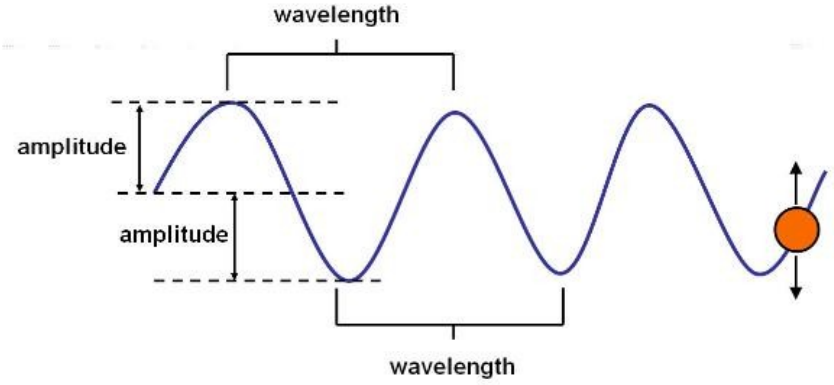
What is wavelength?

_____

Draw a sound wave for:

(Label the following on each diagram: wavelength, amplitude, trough, crest)

A loud sound:	A softer, quieter sound:



Use this diagram to help

What is the Doppler effect? Give an example of when you would experience this.

Describe how ultrasound is different to infrasound.

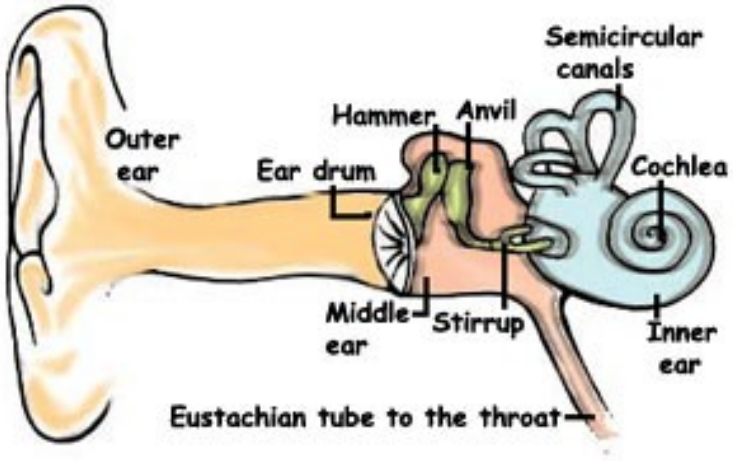
Describe how each of the following instruments produces sound.





The ear collects sound waves, amplify their vibrations and convert these into electrical impulses. Your brain then interprets these impulses as sound. (You will need to add the pinna, oval window and auditory nerve to this diagram below. Note: hammer, anvil and stirrup = the ossicles)

Draw vertical lines to show the outer, middle and inner ear



Eardrum:

_____

_____

_____

Pinna:

_____

_____

_____

Oval Window:

_____

_____

Semicircular canals:

_____

_____

_____

Auditory nerve:

_____

_____

Cochlea:

Eustachian tube:

The Ossicles:

Explain how exposure to high noise levels (>85dB) can permanently damage your hearing. Reference the condition tinnitus within your answer. Note: dB = decibels, measure of ‘loudness’.